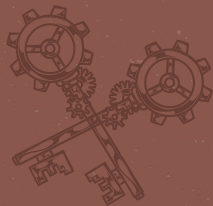

INDUSTRIALIZATION AND INNOVATION

INTRODUCTION

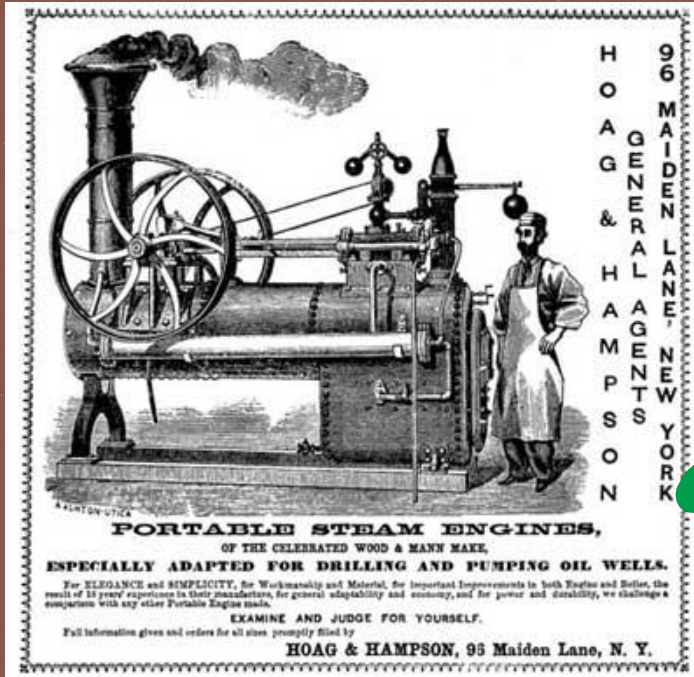


Date	Invention	Invented or Patented by
1851	improved method of converting iron to steel	William Kelly
1852	passenger elevator	Elisha Otis
1856	perfected "Bessemer" process	Henry Bessemer
1858	machine for sewing shoe soles onto uppers	Gordon McKay
1859	first used steam engine to drill for oil	Edwin Drake
1864	sleeping car	George Pullman
1865	compression ice machine	Thaddeus Lowe
1867	typewriter	Christopher Sholes
1868	air brake	George Westinghouse
1869	electric voting machine	Thomas A. Edison
1871	cable streetcar	Andrew S. Hallide
1874	electric streetcar	Stephen Dudley Field
1876	carpet sweeper	Melville Bissell
1876	telephone	Alexander Graham Bell
1878	phonograph	Thomas A. Edison
1879	first practical incandescent light bulb	Thomas A. Edison
1879	cash register	James Ritty
1882	electric iron	Henry W. Seely
1884	fountain pen	Lewis E. Waterman
1886	electric welding machine	Elihu Thomas
1888	safety razor with throwaway blades	King C. Gillette
1888	mass-produced camera	George Eastman
1888	revolving door	Theophilus Van Kannel
1889	electric sewing machine	Singer Manufacturing Co.
1891	escalator	Jesse Reno
1893	gasoline-powered car	Charles and J. Frank Duryea
1893	zip fastener	Whitcomb L. Judson
1899	motor-driven vacuum cleaner	John Thurman





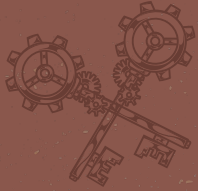
What are the big inventions the text focuses on in 3.1?





Impacts?

01



Steam Engine to Drill
Oil

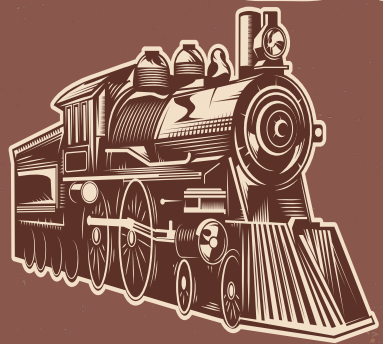


Kerosene



Gasoline





02

Impacts?

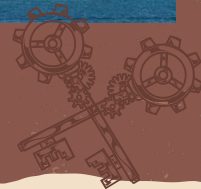
Bessemer Process



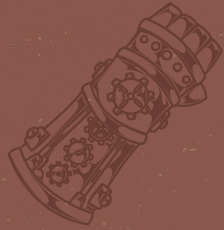
Brooklyn Bridge



Skyscrapers



IMPACTS ON OTHER INVENTIONS?



Date	Invention	Invented or Patented by
1851	improved method of converting iron to steel	William Kelly
1852	passenger elevator	Elisha Otis
1856	perfected "Bessemer" process	Henry Bessemer
1858	machine for sewing shoe soles onto uppers	Gordon McKay
1859	first used steam engine to drill for oil	Edwin Drake
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1891	escalator	Jesse Reno
1993	gasoline-powered car	Charles and J. Frank Duryea
1893	zip fastener	Whitcomb L. Judson
1899	motor-driven vacuum cleaner	John Thurman



IMPACTS ON SOCIETY AND THE ECONOMY?

Big Business and Big Businessmen

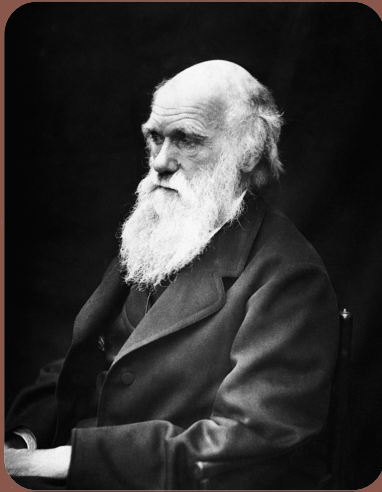
- 4,000 millionaires emerged in the U.S. after the Civil War
- Why now?
 - New uses for natural resources
 - Government support
 - immigration= labor
 - By 1900, 1/3 of all manufacturing was controlled by just 1% of the companies.



- How did these people become millionaires...
 - Social Darwinism
 - New Business Strategies

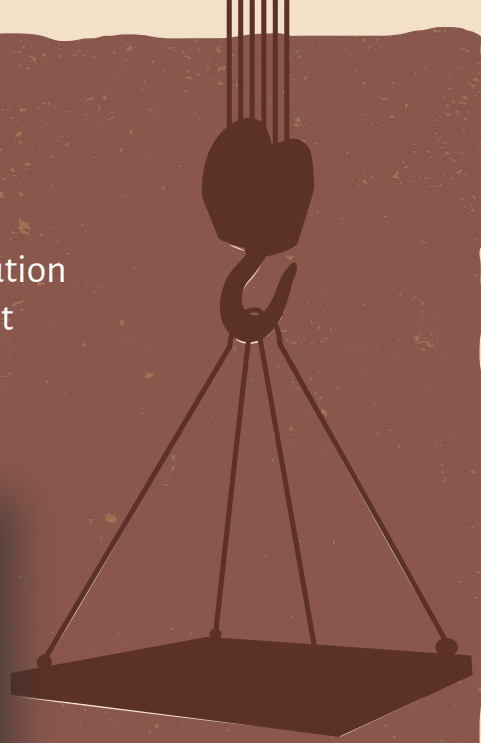
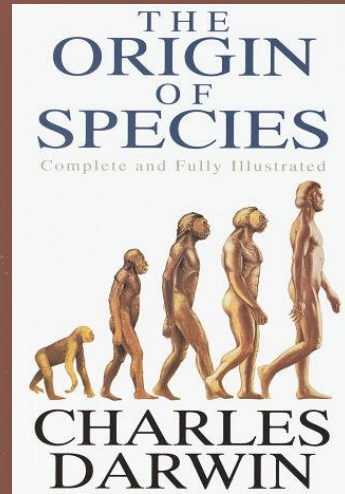
Social Darwinism

Social Darwinism was the application of Charles Darwin's scientific theories of evolution and natural selection to contemporary social development. In nature, only the fittest survived—so too in the marketplace. This form of justification was enthusiastically adopted by many American businessmen as scientific proof of their superiority.

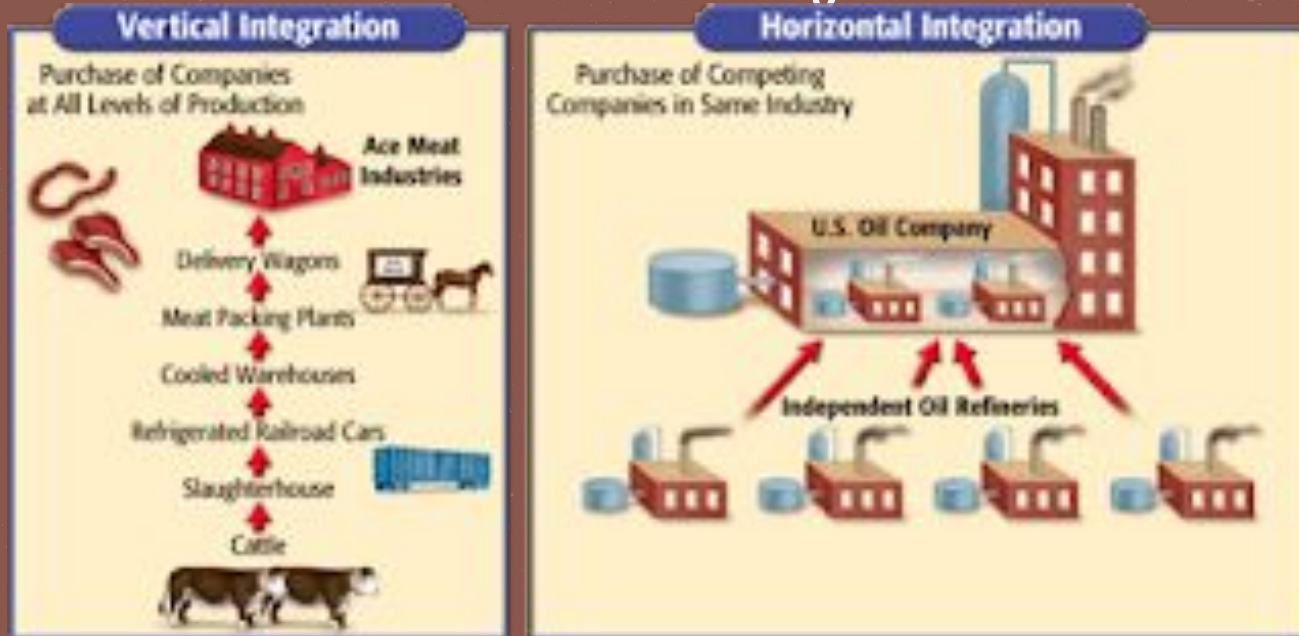


*Why are the
poor, poor?*

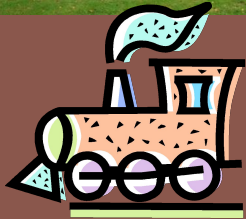
*Why are the
rich, rich?*



New Business Strategies



- Vertical Integration
- Horizontal Integration



Vertical Integration



*This
process
allows you
to cut out
the...*



Horizontal Integration

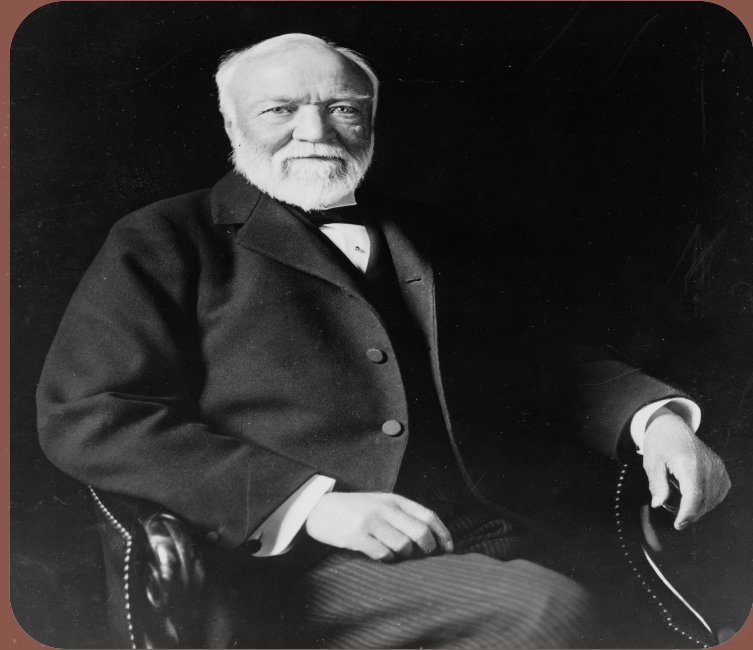


*What if
you
owned
all of
the
looms?*

Examples of these new millionaires



What business strategy did he use?



What business strategy did he use?

Important Terms for this Section

Industrialization

Captain of Industry

Robber Baron

Philanthropist

Vertical and Horizontal Integration

Monopoly

Social Darwinism

Bessemer Process

J.D. Rockefeller

Andrew Carnegie